

SAMPLE COLLEGE OF ENGINEERING

Syllabus extract · B.E. Computer Science · Semester 4
(classroom sample document, not a real syllabus)

CS8401 · Operating Systems

Unit 1 — Processes and threads. Process states: New, Ready, Running, Waiting, Terminated. Context switching and the process control block. Threads share code, data and heap; each thread keeps its own stack and registers. Unit 2 — CPU scheduling. FCFS, SJF, priority and round robin. Turnaround time, waiting time and response time. A very large time quantum degenerates round robin into FCFS; a very small one spends the CPU on context switches. Unit 3 — Deadlock. The four necessary conditions are mutual exclusion, hold and wait, no pre-emption and circular wait. Banker's algorithm and the safe sequence. Unit 4 — Memory. Paging, segmentation, the TLB, and page replacement: FIFO, LRU and OPT. FIFO can produce more page faults with more frames, which is Belady's anomaly. Unit 5 — File systems and I/O.

Assessment pattern

Internal assessment carries 50 marks: two tests of 20 marks scaled to 30, plus 20 marks for assignments and quizzes. The end-semester examination carries 50 marks. To pass a theory course a student needs at least 45% in the end-semester examination and 50% of the total marks in the course. For laboratory courses the split is 60 marks internal and 40 marks end-semester practical, with a minimum of 50% in the practical examination.

Attendance and arrears

A minimum of 75% attendance is required to write the end-semester examination. Attendance between 65% and 75% may be condoned on medical grounds with the approval of the committee. An arrear candidate re-appears for the end-semester examination only; internal marks are retained. More than four standing arrears at the end of Semester 4 blocks registration for higher-semester courses.

Note

This document is written for a classroom exercise. For anything that matters, read the actual regulations for your batch on the college website.